MILITARY SPECIFICATION

FUZE, PD, M536, SEPARATION CHARGE FOR LOADING, ASSEMBLING AND PACKING

1. SCOPE

1.1 This specification covers the loading, assembling and packing for one type of separation charge designated as Fuze, PD, M536, Separation Charge for.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-116	-	Preservation, Methods of
MIL-A-2550	-	Ammunition and Special Weapons, General Specification for
MIL-I-45607	-	Inspection Equipment, Supply and Maintenance of

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)

FSC: 1390

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MIL-STD-109	-	Quality Assurance Terms and Definitions
MIL-STD-1168	-	Lot Numbering of Ammunition
MIL-STD-1235	-	Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

US ARMY MUNITIONS COMMAND

8864622	-	Separation Charge Assembly

PUBLICATIONS

ARMY

ET-8864622	-	Equipment Tabulation

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functiongs should be obtained from the procuring activity or as directed by the Contracting Officer).

2.2 Other publications.-The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

CODE OF FEDERAL REGULATIONS

Title 49	-	Transportation, Parts 171-179

(The Code of Federal Regulations is available from the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402. Orders should specify 49 CFR 171-179 (latest revision).)

3. REQUIREMENTS

3.1 Material.Materials shall be in accordance with the applicable drawings and specifications.

3.2 Parts.-The parts shall comply with all requirements specified on Drawing (Dwg) 8864622, all associated drawings, and with all requirements specified in applicable specifications.

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3.3 Workmanship.-All parts and assemblies shall be fabricated, loaded and assembled in a thorough, workmanlike manner. They shall be free of burrs, sharp edges, cracks, dirt, grease, rust, and other foreign matter. The cleaning method used shall not be injurious to any part, nor shall the parts be contaminated by the cleaning agents. Exterior surface coatings shall be continuous except for a few light scratches not exposing base material.

3.4 Submission of results of contractor-conducted examinations and tests.-The contractor shall forward copies of the results of any of his examinations or tests to the designated address as required by the contract. (See 6.4)

3.5 First article testing.-This specification makes provisions for first article testing. Submission of first article quantity by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 to define terms used herein. The provisions of MIL-A-2550 shall apply.

4.1.1 Submission of product.-At the time the completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all the quality assurance provisions specified in this specification.

b. Drawing and specification number and date, together with identification and date of changes thereto.

c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.

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d. Number of items in the lot.
e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 First article inspection.

4.2.1 Submission.-The contractor shall submit a sample of one hundred five (105) separation charge assemblies to a Government proving ground designated by the Contracting Officer for testing in accordance with the provisions of 4.2.2 (See 6.1). All parts and materials shall be the same as will be used in production and shall be obtained from the same source of supply. Prior to submission, the sample shall be inspected by the contractor to the degree necessary to assure that it conforms to the requirements of the contract.

4.2.2 Inspection to be performed.-Assemblies will be subjected by the Government to the following tests:

4.2.2.1 Seal of separation charge assembly.-The one hundred five (105) separation charge assembles shall be subjected to this test. Seventy-five (75) of these assemblies shall be subjected to exposure as specified on the applicable drawing. The remaining thirty (30) assemblies shall remain unexposed. Each of the 105 assemblies shall be statically functioned.

4.2.3 Rejection.-If the standard deviation (See 6.6) of the exposed group (75) is equal to or less than the standard deviation (See unexposed group (30)) and if the arithmetic mean of the exposed group is equal to or more than the arithmetic mean of the unexposed group, the lot shall be accepted. If the standard deviation of the exposed group is greater than the standard deviation of the unexposed group, the "f" value shall be determined as specified in 6.5.1, or if the arithmetic mean of the exposed group is less than the arithmetic mean of the unexposed group, the "t" value shall be determined as specified in 6.5.2. If the "f" value is equal to or more than 2.19 or the "t" value is equal to or more than 2.36 or one or more of the one hundred five (105) assemblies fail to function, the first article quantity shall be rejected. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit a new first article quantity. Until the first article quantity is accepted, the contractor is not authorized by the Government to proceed with regular production unless otherwise directed by the Contracting Officer.

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4.3 Inspection provisions.

4.3.1 Lot formation.-The term "inspection lot" as used in this specification is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision and complies with the provisions for submission of product as specified in MIL-STD-105. Changes to process, specification, or drawing not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168. Each inspection lot of separation charge assemblies shall contain:

a. Metal parts from one lot interfix from one manufacturer.

b. Propellant from one lot number from one manufacturer.

4.3.2 Examination.-The examinations listed in the following subparagraphs shall be performed on inspection lots (See 4.3.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for critical defects (and major defects, when so specified) shall be 100 percent. Sampling plans for major and minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the AQL's designated herein to the groups of major and minor defects listed, the contractor may elect to apply AQL's to individual defects using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. When this method is elected, it shall be applied to all of the major and minor defects listed in the applicable subparagraph of this section, except where 100 percent inspection is specified, and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

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4.3.2.1 Separation charge assembly, prior to crimping (See Dwg 8864622).-

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			
Major	AQL 0.40 Percent		
101.	Weight of propellent	Balance	01001
Minor: None defined			

4.3.2.2 Separation charge assembly (See Dwg 8864622).-

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			
Major	AQL 0.65 Percent		
101.	Height, max	Gage	02001
102.	Diameter, max	Gage	02002
103.	Crimp incomplete	Visual	02003
104.	Any component damaged (puncture or cut)	Visual	02004
Minor	AQL 0.65 Percent		
201.	Evidence of poor workmanship (See 3.3)	Visual	02005

4.3.2.3 Unsealed fiber box (See Dwg 9252340).-

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			
Major	AQL 0.40 Percent		
101.	Improper number of separation charge assemblies	Visual	03001
102.	Fillers missing	Visual	03002
Minor: None defined			

4.3.2.4 Sealed fiber box, prior to sealing bag (See Dwg 9252340).-

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			

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Major:	AQL 0.40 Percent		
101.	Container damaged	Visual	04001
102.	Corners not blunted	Visual	04002
Minor:	AQL		
201.	Contents loose	Manual	04003
202.	Marking misleading or unidentifiable	Visual	04004

4.3.2.5 Unsealed wooden packing box (See Dwg. 9252341).-

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.65 Percent		
101.	Incorrect number of cartons	Visual	05001
102.	Bag torn or perforated	Visual	05002
103.	Bag improperly sealed	Visual	05003
Minor:	None defined		

4.3.2.6 Sealed wooden packing box (See Dwg. 9252341).-

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 Percent		
101.	Box damaged	Visual	06001
Minor:	AQL 1.50 Percent		
201.	Contents loose	Manual	06002
202.	Seal missing or improperly assembled	Visual	06003
203.	Marking misleading or unidentifiable	Visual	06004

4.3.3 Testing.-

4.3.3.1 Seal of separation charge assembly (output) (See Dwg. 8864622), Major defect, Code No. 07001.-At the loading station, after sealant is applied, one hundred five (105) separation charge assemblies shall be selected from each lot and subjected to this test. Seventy-five (75) of the assemblies shall be subjected to exposure as specified on the applicable drawing. Each

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of the one hundred five (105) assemblies shall be statically functioned. If the standard deviation (See 6.5) of the exposed group (75) is equal to or less than the standard deviation, (See unexposed group (30))and if the arithmetic mean of the unexposed grcup, the lot shall be accepted. If the standard deviation of the exposed group is greater than the standard deviation of the unexposed group, the "f" value shall be determined as specified in 6.5.1, or if the arithmetic mean of the exposed group is less then the arithmetic mean of the unexposed, the "t" value shall be determined as specified in 6.5.2. If the "f" value is equal to or more than 2.19 or the "t" value is equal to or more than 2.36 or one or more of the one hundred five (105) assemblies fail to function, the lot shall be rejected. This test shall be performed as specified in 4.4.1, using equipment in accordance with 4.3.4.

4.3.3.2 Heat seal test of packing seals (See Dwg. 9252340), Major defect, Code No. 08001.-The samples shall be selected in accordance with MIL-P-116. Rejection shall be in accordance with MIL-P-116. The test shall be performed as specified in 4.4.2.

4.3.4 Inspection equipment.-Equipment Tabulation (ET) number 8864622 and Equipment Lists (EL's) referenced thereon identify the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in 6.3. Supply and maintenance of the equipment listed on the EL's shall be in accordance with MIL-I-45607.

4.3.4.1 Government rights to documentation.-Inspection equipment drawings and lists provided and revised in accordance with the requirements of this specification may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacements parts, inspection of items at overhaul, general maintenance of equipment, construction, survey, and whenever inspection equipment drawings are needed.

4.4 Test methods and procedures.

4.4.1 Seal of separation charge assembly (output).-The separation charge assembly shall be statically functioned, utilizing the equipment specified on the Equipment List.

4.4.2 Heat seal test of packing seals.-This test shall be performed in accordance with MIL-P-116. Seals so tested shall not be returned to the lot.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

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5.1.1 Level C.-The separation charge assembly shall be preserved and packaged in accordance with Dwg. 9252340.

5.2 Packing

5.2.1 Level C.-The separation charge assembly, preserved and packaged as specified in 5.1.1, shall be packed for shipment in accordance with Dwg. 9252341.

5.3 Marking.-Marking shall be in accordance with Dwgs. 9252340 and 9252341 and Code of Federal Regulations, Title 49 Parts 171-179.

5.4 Shipping.-When shipments from more than one lot are shipped as a carload, each lot shall be kept separate, and the division between lots clearly indicated to prevent mixing of the lots in transit.

6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.
b. Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.
c. Provisions for submission of first article samples.

6.2 Inspection code numbers.-The five-digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government. These code numbers are also used to correlate the characteristics cited on Equipment Lists with the inspections listed in this specification. These code numbers shall be listed on drawings of equipment designs submitted by the contractor (see 6.3.5).

6.3 Inspection equipment.-The contractor shall design inspection equipment as required by the referenced Equipment Lists (EL) in accordance with the instructions of paragraphs 6.3.1 through 6.3.5.

6.3.1 Equipment lists.-Equipment lists indicate the availability of inspection equipment designs by showing in the "Part Number" column of the list of inspection equipment (Form SMUPA1010) the numbers of drawings or Federal Stock numbers of existing equipment designs, or codes as indicated in paragraph 6.3.2. Design action required of the contractor is described in paragraphs 6.3.3 and 6.3.4. The contractor will be required to prepare detailed drawings in accordance with 6.3.4 for all the equipment coded as "Contractor Design" in the "Part Number" column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

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6.3.2 Equipment list codes.-The inspection equipment as defined in 6.3.3 and 6.3.4 will be designated in the EL by the following codes:

CD - Contractor design on controlled contractor format and/or commercial equipment.
MU - Army design, mandatory for use.

6.3.3 Army designs.-Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications: Mandatory (designated "MU") and non-mandatory (indicated by drawing or Federal Stock Number). When an equipment list referenced mandatory Army Designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory design, with the approval of the Government, design alternate inspection equipment to facilitate his operations. Such contractor-prepared designs (see 6.3.4) or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.4 Contractor designs.-Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finishes, notes and operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Unless otherwise specified, contractor designs may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above, subject to the following controls: All submitted contractor designs shall conform to Specification MIL-D-1000, Category E, Form 2. Legibility and reproducibility shall permit conventional making of clearly understandable, high-contrast reproductions. Contractors shall submit three copies of final designs as a flat set. Designs shall be submitted for approval as specified in 6.3.5.

6.3.5 Submission of contractor designs.-All submitted designs shall contain a reference to the applicable 5-digit code number contained in Section 4 of this specification and the appropriate component or assembly drawing number and revision letter to which the specification designs applied.

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Unless otherwise specified on the EL, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND13. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract. Submission shall be as directed by the contracting officer. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of design.

6.4 Submission of results of contractor-conducted examinations and tests.-Unless otherwise specified by the Contracting Officer, the contractor shall forward requested records of examinations or tests to Commanding Officer, Picatinny Arsenal, ATTN" SMUPA-ND3.

6.5 Standard deviation shall be computed as follows:

$$\sigma = \sqrt{\frac{\Sigma X^2 \quad (\Sigma X)^2}{n - 1}}$$

where:

σ = standard deviation
Σ = sum of
X = differences between individual value and average n - 1 = number of observation minus one

6.5.1 (X +1) The "f" value shall be determined as follows:

$$f = \frac{\sigma e^2}{\sigma n2}$$

where:

σe^2 = standard deviation squared of the exposed group

σn^2 = standard deviation squared of the unexposed group.

6.5.2 (X + 2) The "t" value shall be determined as follows:

$$t = \frac{\bar{X}n - \bar{X}e}{t} \quad X\ 4.63$$

where:

$\bar{X}n$ = arithmetic mean of the unexposed group
$\bar{X}e$ = arithmetic mean of the exposed group
t = standard deviation of total

The standard deviation of the total shall be determined as follows:

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$$\sigma t = \frac{30\,\sigma n^2 + 75\,\sigma e^2}{105}$$

Custodian:
Army-MU

Preparing activity:
Army-MU

Project Number: 1390-A-790

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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